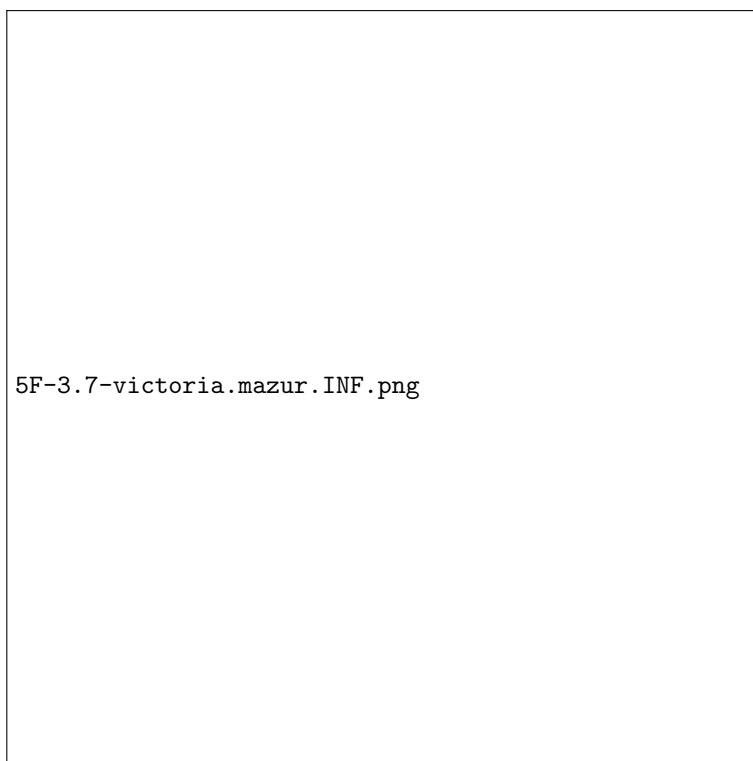


Auteur: Victoria Mazur
Studierichting: Informatica
Oefeningenreeks 5F nr. 3.7

$$\begin{aligned} I &= \frac{1}{2} \int_0^\pi (1 + \cos \theta)^2 d\theta - \frac{1}{2} \int_0^{\frac{\pi}{2}} (\cos \theta)^2 d\theta \\ &= \frac{1}{2} \int_0^\pi (1 + 2 \cos \theta + \cos^2 \theta) d\theta - \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \\ &= \frac{1}{2} [\theta]_0^\pi + [\sin \theta]_0^\pi + \frac{1}{2} \int_0^\pi \cos^2 \theta d\theta - \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \\ &= \frac{\pi}{2} + 0 + \frac{1}{2} \int_0^\pi \cos^2 \theta d\theta - \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \\ &= \frac{\pi}{2} + \frac{1}{2} \int_0^\pi \left(\frac{\cos 2\theta + 1}{2} \right) d\theta - \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{\cos 2\theta + 1}{2} \right) d\theta \\ &= \frac{\pi}{2} + \frac{1}{4} \int_0^\pi (\cos 2\theta + 1) d\theta \\ &= \frac{\pi}{2} + \frac{1}{4} \int_{\frac{\pi}{2}}^\pi \cos 2\theta d\theta + \frac{1}{4} \int_{\frac{\pi}{2}}^\pi 1 d\theta \\ &= \frac{\pi}{2} + \frac{1}{8} [\sin 2\theta]_{\frac{\pi}{2}}^\pi + \frac{\pi}{8} \\ &= \frac{\pi}{2} + \frac{\pi}{8} = \frac{5\pi}{8} \end{aligned}$$

References



5F-3.7-victoria.mazur.INF.png